

Hence we conclude that there exists $\sigma : [0, \ell_i] \rightarrow \mathbb{R}^2$ such that

$$x(s) = x(0) + \frac{n_1}{\ell_i}s + \sigma_1(s), \quad y(s) = y(0) + \frac{n_2}{\ell_i}s + \sigma_2(s),$$

and moreover

$$(2.6) \quad \|\sigma(s)\|_\infty + \|\sigma'(s)\|_\infty \leq C\varepsilon_0.$$

By the implicit function theorem (see also [22, Lemma 3.4]), the curve Γ_i can be parametrized as a graph over the line L_i with slope n_1/n_2 (where L_i is a line parallel to $x = 0$ when $n_2 = 0$) and passing through $(x(0), y(0))$ with height function $f_i \in C^{1, \frac{1}{2}}(L_i)$ with

$$(2.7) \quad \|f_i\|_{C^1(L_i)} \leq C\varepsilon_0.$$

Since Γ_i does not bound, there exists a unique $\tilde{\Gamma}_i$ such that $\Gamma_i \cup \tilde{\Gamma}_i$ is the boundary of a connected component E_i of E . By the previous arguments, also $\tilde{\Gamma}_i$ is a small $C^{1, \frac{1}{2}}$ -deformation of a line $\tilde{L}_i$ parallel to L_i . By parallel shifting the curves Γ_i and $\tilde{\Gamma}_i$, we can find a strip S_i with boundary $L_i \cup \tilde{L}_i$ and $|E_i| = |S_i|$. Thanks to (2.6), (2.7) still holds, possibly increasing C . By the area formula, it is easy to see that (see e.g. [7, Lemma 3.1])

$$0 \leq P(E_i) - P(S_i) \leq C\|f_i\|_{C^1(L_i)}^2 \leq C\varepsilon_0^2 = C\|\kappa_E - \bar{\kappa}_E\|_{L^2(\partial E)}^2.$$

We can then sum over all the connected components (whose number is bounded by $M/2$) to get the desired result. This completes the proof. $\square$

3. ASYMPTOTIC BEHAVIOR OF THE MULLINS-SEKERKA FLOW

In this section we apply the quantitative Alexandrov Theorem (Theorem 1.2) to show the asymptotic of the Mullins-Sekerka flow. We start by recalling the definition of *flat flow* solutions of the Mullins-Sekerka flow, as given in [25, 37].

Let $E \subseteq \mathbb{T}^2$ be a measurable set with $|E| = m$, and consider the minimization problem

$$(3.1) \quad \min \left\{ P(F) + \frac{h}{2} \int_{\mathbb{T}^2} |\nabla U_{F,E}|^2 dx : |F| = |E| \right\},$$

where $U_{F,E} \in H^1(\mathbb{T}^2)$ is the solution to

$$(3.2) \quad -\Delta U_{F,E} = \frac{1}{h} (\chi_F - \chi_E) \quad \text{in } \mathbb{T}^2$$

with zero average. Let us set

$$(3.3) \quad \mathcal{D}(F, E) := \int_{\mathbb{T}^2} |\nabla U_{F,E}|^2 dx.$$

By the results of [25, 37], there exists a minimizer of (3.1), which may not be unique. Note also that, defining the H^{-1} -norm by duality as

$$\|f\|_{H^{-1}(\mathbb{T}^2)} := \sup \left\{ \int_{\mathbb{T}^2} \varphi f dx : \|\nabla \varphi\|_{L^2(\mathbb{T}^2)} \leq 1 \right\},$$

pairing this definition with (3.2) and integrating by parts, we get

$$(3.4) \quad \|\chi_F - \chi_E\|_{H^{-1}(\mathbb{T}^2)}^2 \leq h^2 \|\nabla U_{F,E}\|_{L^2(\mathbb{T}^2)}^2 = h^2 \mathcal{D}(F, E).$$